Trigonometric identities:

$$
\begin{aligned}
e^{j\theta} &= \cos(\theta) + j\sin(\theta), \\
\cos(\theta) &= \frac{1}{2}[e^{j\theta} + e^{-j\theta}], \\
\sin(\theta) &= \frac{1}{2j}[e^{j\theta} - e^{-j\theta}], \\
2\cos(\theta)\cos(\phi) &= \cos(\theta+\phi) + \cos(\theta-\phi), \\
2\sin(\theta)\sin(\phi) &= \cos(\theta-\phi) - \cos(\theta+\phi), \\
2\sin(\theta)\cos(\phi) &= \sin(\theta+\phi) + \sin(\theta-\phi), \\
\cos(\theta+\phi) &= \cos(\theta)\cos(\phi) - \sin(\theta)\sin(\phi), \\
\cos(\theta-\phi) &= \cos(\theta)\cos(\phi) + \sin(\theta)\sin(\phi), \\
\sin(\theta+\phi) &= \sin(\theta)\cos(\phi) + \cos(\theta)\sin(\phi), \\
\sin(\theta-\phi) &= \sin(\theta)\cos(\phi) - \cos(\theta)\sin(\phi).
\end{aligned}
$$

Fourier transform of special functions:

$$
\begin{aligned}
\mathrm{rect}(t) &\xleftrightarrow{\mathcal{F}} \mathrm{sinc}(f), \\
\mathrm{triang}(t) &\xleftrightarrow{\mathcal{F}} \mathrm{sinc}^2(f),
\end{aligned}
$$

where

$$
\begin{aligned}
\mathrm{rect}(t) &= \begin{cases} 1, & |t| < \frac{1}{2}, \\ 0, & \text{otherwise}, \end{cases} \\
\mathrm{triang}(t) &= \begin{cases} 1-|t|, & |t| < 1, \\ 0, & \text{otherwise}, \end{cases} \\
\mathrm{sinc}(t) &= \frac{\sin(\pi t)}{\pi t}.
\end{aligned}
$$

Properties of Fourier transform:

If $g(t) \xleftrightarrow{\mathcal{F}} G(f)$ and $h(t) \xleftrightarrow{\mathcal{F}} H(f)$, then

$$
\begin{aligned}
g(at) &\xleftrightarrow{\mathcal{F}} \frac{1}{|a|}G\left(\frac{f}{a}\right), \\
G(t) &\xleftrightarrow{\mathcal{F}} g(-f), \\
g(t-t_0) &\xleftrightarrow{\mathcal{F}} G(f)e^{-j2\pi f t_0},
\end{aligned}
$$

$$g(t)e^{j2\pi f_c t} \stackrel{\mathcal{F}}{\longleftrightarrow} G(f - f_c),$$

$$g(t)\cos(2\pi f_c t) \stackrel{\mathcal{F}}{\longleftrightarrow} \frac{1}{2}\left[G(f - f_c) + G(f + f_c)\right],$$

$$g(t) * h(t) \stackrel{\mathcal{F}}{\longleftrightarrow} G(f)H(f),$$

$$g(t)h(t) \stackrel{\mathcal{F}}{\longleftrightarrow} G(f) * H(f),$$

$$\int_{-\infty}^{\infty} |g(t)|^2 \mathrm{d}t = \int_{-\infty}^{\infty} |G(f)|^2 \mathrm{d}f.$$

Amplitude modulation:

Standard AM: $s(t) = A_c[1 + k_a m(t)]\cos(2\pi f_c t)$.

DSB-SC: $s(t) = A_c m(t)\cos(2\pi f_c t)$.

Mean power of a sinusoid:

The mean power of $A\sin(2\pi f t + \theta)$ is $\frac{A^2}{2}$.

Autocorrelation function and power spectral density function:

Let $X(t)$ be a wide-sense stationary random process and $Y(t) = h(t) * X(t)$, where $h(t)$ is the impulse response of a linear time-invariant filter.

$$R_X(\tau) = \mathbb{E}[X(t)X(t+\tau)], \qquad R_Y(\tau) = \mathbb{E}[Y(t)Y(t+\tau)],$$

$$R_X(\tau) \stackrel{\mathcal{F}}{\longleftrightarrow} S_X(f), \qquad R_Y(\tau) \stackrel{\mathcal{F}}{\longleftrightarrow} S_Y(f),$$

$$\mathbb{E}[X^2(t)] = R_X(0) = \int_{-\infty}^{\infty} S_X(f)\mathrm{d}f, \qquad \mathbb{E}[Y^2(t)] = R_Y(0) = \int_{-\infty}^{\infty} S_Y(f)\mathrm{d}f,$$

$$S_Y(f) = |H(f)|^2 S_X(f) \text{ with } h(t) \stackrel{\mathcal{F}}{\longleftrightarrow} H(f).$$